

Qualification Pack



Sports Bat and Racquet Craftsperson (Junior)

QP Code: SPF/Q8102

Version: 1.0

NSQF Level: 3

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SPF/Q8102: Sports Bat and Racquet Craftsperson (Junior)

Brief Job Description

Sports bat and racquet craftsperson are involved in production, assembling, and finishing sports bats and racquets such as cricket bats, baseball bats, table tennis, lawn tennis, badminton racquets, etc. Their tasks range from operating machines to fine handcrafting of equipment

Personal Attributes

Individuals in this role should be creative, details-oriented, self-motivated and energetic. Individuals in this role must demonstrate a very good listening and comprehension skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SPF/N8103: Prepare tools and materials to make bats and racquets](#)
2. [SPF/N8104: Craft bats and racquets](#)
3. [SPF/N8105: Carry out final steps of crafting the bats and racquets](#)
4. [SPF/N1169: Improve workplace resource usage](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Sports
Sub-Sector	Sports Manufacturing
Occupation	Sports Manufacturing
Country	India
NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO

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Minimum Educational Qualification & Experience	10th grade pass OR 9th grade pass with 1 Year of experience OR 8th grade pass with 2 Years of experience OR 5th grade pass with 5 Years of experience OR Certificate-NSQF (Level2-Sports Goods Manufacturing Helper) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	23/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-03-SP-00607-2023-V1-SPEFLSC
NQR Version	1

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SPF/N8103: Prepare tools and materials to make bats and racquets

Description

This unit is about preparing equipment and materials for production of sports goods.

Scope

The scope covers the following :

- Select equipment for the production of bats and racquet
- Prepare materials for the production bats and racquet
- Maintain hygiene and sanitation

Elements and Performance Criteria

Select tools and machinery for crafting sports bats

To be competent, the user/individual on the job must be able to:

- PC1.** ensure the availability of all required equipment and tools for the production
- PC2.** use the correct tools and equipment
- PC3.** check that equipment is safe and set up in readiness for use
- PC4.** select the correct component parts for the style being worked on
- PC5.** conform to company quality standards
- PC6.** report any damaged equipment to the responsible person
- PC7.** follow company reporting procedures about defective tools and machines which affect work and report risks/ problems
- PC8.** sort and place work to assist the next stage of production and minimize the risk of damage

Prepare materials for the production

To be competent, the user/individual on the job must be able to:

- PC9.** identify the properties of wood
- PC10.** select type of wood based on the product to be manufactured
- PC11.** select the wood that can be used in making of cricket bats, baseball bats, table tennis racquets, etc.
- PC12.** identify the process of grading wood
- PC13.** select types of tools and equipment that should be used for shaping the wood
- PC14.** ensure correct wood is selected before it is put in use
- PC15.** check quality of wood and maintain its standards
- PC16.** identify the process of drying of wood

Maintain hygiene and sanitation

To be competent, the user/individual on the job must be able to:

- PC17.** ensure personal hygiene
- PC18.** ensure equipment, workshop, etc. are sanitized before and after the usage
- PC19.** guide others about hygiene and sanitation workplace requirements

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PC20. report advanced hygiene and sanitation issues to appropriate authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different hand tools used for different leather goods & their cares.
- KU2.** different types of cutting Tools, stitching awls, stitch components.
- KU3.** leather quality and properties
- KU4.** plating and evaluations of leather quality and appropriate uses
- KU5.** types of needles & scissors, knives etc. & their utility
- KU6.** measuring and layout for different shape & sizes of different leather goods
- KU7.** different machines used in manufacturing of leather goods
- KU8.** basic concept of leather goods, kinds, uses, names, functions of hand tools and other holding devices
- KU9.** use and function of different measuring instruments and gauges
- KU10.** process of leather treatment
- KU11.** administration of basic first aid

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and understand all usage and safety manuals
- GS2.** communicate in a clear and polite manner
- GS3.** comprehend instructions
- GS4.** instruct users with clear and easy to understand instructions
- GS5.** communicate effectively with senior citizens and young children to maintain safety
- GS6.** assess tools and equipment conditions
- GS7.** convert metrics of measurements (feet, meter, millimeter, etc.)

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Select tools and machinery for crafting sports bats</i>	8	32	-	-
PC1. ensure the availability of all required equipment and tools for the production	1	4	-	-
PC2. use the correct tools and equipment	1	4	-	-
PC3. check that equipment is safe and set up in readiness for use	1	4	-	-
PC4. select the correct component parts for the style being worked on	1	4	-	-
PC5. conform to company quality standards	1	4	-	-
PC6. report any damaged equipment to the responsible person	1	4	-	-
PC7. follow company reporting procedures about defective tools and machines which affect work and report risks/ problems	1	4	-	-
PC8. sort and place work to assist the next stage of production and minimize the risk of damage	1	4	-	-
<i>Prepare materials for the production</i>	8	32	-	-
PC9. identify the properties of wood	1	4	-	-
PC10. select type of wood based on the product to be manufactured	1	4	-	-
PC11. select the wood that can be used in making of cricket bats, baseball bats, table tennis racquets, etc.	1	4	-	-
PC12. identify the process of grading wood	1	4	-	-
PC13. select types of tools and equipment that should be used for shaping the wood	1	4	-	-
PC14. ensure correct wood is selected before it is put in use	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. check quality of wood and maintain its standards	1	4	-	-
PC16. identify the process of drying of wood	1	4	-	-
<i>Maintain hygiene and sanitation</i>	4	16	-	-
PC17. ensure personal hygiene	1	4	-	-
PC18. ensure equipment, workshop, etc. are sanitized before and after the usage	1	4	-	-
PC19. guide others about hygiene and sanitation workplace requirements	1	4	-	-
PC20. report advanced hygiene and sanitation issues to appropriate authority	1	4	-	-
NOS Total	20	80	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SPF/N8103
NOS Name	Prepare tools and materials to make bats and racquets
Sector	Sports
Sub-Sector	Sports Manufacturing
Occupation	Sports Manufacturing
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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SPF/N8104: Craft bats and racquets

Description

This unit is about performing cutting operations using hand and machine.

Scope

The scope covers the following :

- Craft cricket bats
- Craft table tennis racquets
- Assemble lawn tennis racquets

Elements and Performance Criteria

Craft cricket bats

To be competent, the user/individual on the job must be able to:

- PC1.** calculate the number of components needed for production
- PC2.** select appropriate willow to craft bats
- PC3.** identify the properties of Kashmir willow, English willow, and White willow
- PC4.** ensure the willows are free from moisture before using them
- PC5.** cut or match the willow to create clefts
- PC6.** use waxing technique to smoothen the edges
- PC7.** grade clefts as per four levels based on straightness, width, blemishes
- PC8.** ensure the standard size of the bats are 38 inches and width as 4.25 inches
- PC9.** use pressing machine to compress the willow to a solid compact state
- PC10.** cut/slice the willow or the blade to create space for the handle
- PC11.** assemble handle with the blades using wood adhesive
- PC12.** use drawknife to shape the shoulders of the blades to the cricket bat
- PC13.** ensure rounding off the tow and filling away unnecessary pieces on the bat
- PC14.** use sanding process to smoothen the surface of the bat
- PC15.** operate foot treadle to bind the handle of the bats with twine
- PC16.** transfer bats for the quality check and next stage of production

Craft table tennis racquets

To be competent, the user/individual on the job must be able to:

- PC17.** identify the types of laminated woods to be used to craft table tennis racquet
- PC18.** dry the wood using sun drying techniques or an incubator
- PC19.** measure the dryness of the wood using appropriate tool
- PC20.** identify the properties of plywood cypress, wood balsa, soft ash, walnut, etc.
- PC21.** select appropriate laminated wood to craft table tennis racquet
- PC22.** slice the wood into core blades

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- PC23.** ensure uniformity of all sliced blades – shape and size
- PC24.** calculate the number of components needed for production
- PC25.** ensure the rubber and sponge used as the surface of the racquet is ITTF approved (International Table Tennis Federation)
- PC26.** identify the implications of using inward and outward pointing pimples rubber on the surface of the racquet
- PC27.** apply sponge and rubber to the wood using adhesive
- PC28.** utilise rubber pieces efficiently to get maximum cut pieces with minimum wastage
- PC29.** ensure no defects on the material
- PC30.** check the stretchability and tightness of the rubber as per the ITTF standards
- PC31.** transfer racquets for the quality check and next stage of production
- PC32.** avoid any imperfections in the material when assembling
- PC33.** dispose of waste materials safely and return re-useable materials
- PC34.** transfer bats for the quality check and next stage of production

Craft badminton and lawn tennis racquets

To be competent, the user/individual on the job must be able to:

- PC35.** select the material aluminum or composite depending on the frame type and specification
- PC36.** assemble composite materials as 'flat sandwich' (graphite, fiberglass, boron, Kevlar, etc.)
- PC37.** cut layered materials into strips
- PC38.** roll the strips to hollow, flexible tube to create frame
- PC39.** bind layers of sandwiched materials using air-pumped mold or fill the hollow with polyurethane foam
- PC40.** create yoke using drilling machine for string holes
- PC41.** use multiple spindles to hold drill bits to create multiple string holes
- PC42.** use sander to smoothen the rough surface created during drilling
- PC43.** temper the frame using heat and rapid cooling process
- PC44.** anodize the frame by immersing in a mild sulfuric acid solution, and passing through an electric current
- PC45.** ensure the treatment changes the surface of the aluminium, and gives the rackets a shiny finish
- PC46.** fix pre-drilled flexible grommet strip in the groove around the edge of the head
- PC47.** ensure grommet strip fit over the string holes in the frame head

Reporting and Documentation

To be competent, the user/individual on the job must be able to:

- PC48.** identify and respond to imperfections, defects and damage due to mishandling
- PC49.** report defects in the tools and equipment one does not have the authority to repair
- PC50.** report risks/ problems likely to affect services to the relevant person promptly and accurately
- PC51.** complete forms, records and other documentation
- PC52.** make sure that the cut material is correctly sorted to assist the next stage of production
- PC53.** make sure that the cut material carefully placed to minimise the risk of damage
- PC54.** carry out closedown procedures on completion of work

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of willow suitable for crafting bats
- KU2.** types of materials aluminum, composite used for crafting racquets
- KU3.** characteristics of wood, moisture, density, etc.
- KU4.** operating procedures to carry out colouring, cleaning, lamping, thread trimming, and packaging
- KU5.** styles/ design of the products currently being produced
- KU6.** method of interpreting product specifications
- KU7.** method of safely setting up and closing down cutting and other equipment in the work station
- KU8.** equipment needed for the moulding, shaping, sanding, cutting activity
- KU9.** methods of calculating the number of components required for production
- KU10.** common quality imperfections associated with the materials
- KU11.** implications of using defective tools and machines on the materials
- KU12.** manufacturers' instructions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** monitor client behaviour and activities
- GS2.** resolve conflicts
- GS3.** write in English/ local language as applicable
- GS4.** fill up appropriate technical forms, process charts, activity logs in the prescribed format of the company
- GS5.** read English/ local language as applicable
- GS6.** read and understand manuals, health and safety instructions, memos

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Craft cricket bats</i>	16	48	-	-
PC1. calculate the number of components needed for production	1	3	-	-
PC2. select appropriate willow to craft bats	1	3	-	-
PC3. identify the properties of Kashmir willow, English willow, and White willow	1	3	-	-
PC4. ensure the willows are free from moisture before using them	1	3	-	-
PC5. cut or match the willow to create clefts	1	3	-	-
PC6. use waxing technique to smoothen the edges	1	3	-	-
PC7. grade clefts as per four levels based on straightness, width, blemishes	1	3	-	-
PC8. ensure the standard size of the bats are 38inches and width as 4.25 inches	1	3	-	-
PC9. use pressing machine to compress the willow to a solid compact state	1	3	-	-
PC10. cut/slice the willow or the blade to create space for the handle	1	3	-	-
PC11. assemble handle with the blades using wood adhesive	1	3	-	-
PC12. use drawknife to shape the shoulders of the blades to the cricket bat	1	3	-	-
PC13. ensure rounding off the tow and filling away unnecessary pieces on the bat	1	3	-	-
PC14. use sanding process to smoothen the surface of the bat	1	3	-	-
PC15. operate foot treadle to bind the handle of the bats with twine	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. transfer bats for the quality check and next stage of production	1	3	-	-
<i>Craft table tennis racquets</i>	18	54	-	-
PC17. identify the types of laminated woods to be used to craft table tennis racquet	1	3	-	-
PC18. dry the wood using sun drying techniques or an incubator	1	3	-	-
PC19. measure the dryness of the wood using appropriate tool	1	3	-	-
PC20. identify the properties of plywood cypress, wood balsa, soft ash, walnut, etc.	1	3	-	-
PC21. select appropriate laminated wood to craft table tennis racquet	1	3	-	-
PC22. slice the wood into core blades	1	3	-	-
PC23. ensure uniformity of all sliced blades - shape and size	1	3	-	-
PC24. calculate the number of components needed for production	1	3	-	-
PC25. ensure the rubber and sponge used as the surface of the racquet is ITTF approved (International Table Tennis Federation)	1	3	-	-
PC26. identify the implications of using inward and outward pointing pimples rubber on the surface of the racquet	1	3	-	-
PC27. apply sponge and rubber to the wood using adhesive	1	3	-	-
PC28. utilise rubber pieces efficiently to get maximum cut pieces with minimum wastage	1	3	-	-
PC29. ensure no defects on the material	1	3	-	-
PC30. check the stretchability and tightness of the rubber as per the ITTF standards	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. transfer racquets for the quality check and next stage of production	1	3	-	-
PC32. avoid any imperfections in the material when assembling	1	3	-	-
PC33. dispose of waste materials safely and return re-useable materials	1	3	-	-
PC34. transfer bats for the quality check and next stage of production	1	3	-	-
<i>Craft badminton and lawn tennis racquets</i>	13	30	-	-
PC35. select the material aluminum or composite depending on the frame type and specification	1	3	-	-
PC36. assemble composite materials as 'flat sandwich' (graphite, fiberglass, boron, Kevlar, etc.)	1	3	-	-
PC37. cut layered materials into strips	1	3	-	-
PC38. roll the strips to hollow, flexible tube to create frame	1	3	-	-
PC39. bind layers of sandwiched materials using air-pumped mold or fill the hollow with polyurethane foam	1	2	-	-
PC40. create yoke using drilling machine for string holes	1	2	-	-
PC41. use multiple spindles to hold drill bits to create multiple string holes	1	2	-	-
PC42. use sander to smoothen the rough surface created during drilling	1	2	-	-
PC43. temper the frame using heat and rapid cooling process	1	2	-	-
PC44. anodize the frame by immersing in a mild sulfuric acid solution, and passing through an electric current	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC45. ensure the treatment changes the surface of the aluminium, and gives the rackets a shiny finish	1	2	-	-
PC46. fix pre-drilled flexible grommet strip in the groove around the edge of the head	1	2	-	-
PC47. ensure grommet strip fit over the string holes in the frame head	1	2	-	-
<i>Reporting and Documentation</i>	7	14	-	-
PC48. identify and respond to imperfections, defects and damage due to mishandling	1	2	-	-
PC49. report defects in the tools and equipment one does not have the authority to repair	1	2	-	-
PC50. report risks/ problems likely to affect services to the relevant person promptly and accurately	1	2	-	-
PC51. complete forms, records and other documentation	1	2	-	-
PC52. make sure that the cut material is correctly sorted to assist the next stage of production	1	2	-	-
PC53. make sure that the cut material carefully placed to minimise the risk of damage	1	2	-	-
PC54. carry out closedown procedures on completion of work	1	2	-	-
NOS Total	54	146	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SPF/N8104
NOS Name	Craft bats and racquets
Sector	Sports
Sub-Sector	Sports Manufacturing
Occupation	Sports Manufacturing
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	NA
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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SPF/N8105: Carry out final steps of crafting the bats and racquets

Description

This unit is about performing final assembly to produce sports goods

Scope

The scope covers the following :

- Assemble different components of bats
- String and apply finishing touch to racquets
- Perform quality and standards check of bats and racquets
- Work effectively with others

Elements and Performance Criteria

Assemble different components of bats

To be competent, the user/individual on the job must be able to:

- PC1.** identify the points of inspection of the assembling goods
- PC2.** rasp the handles using appropriate tools
- PC3.** ensure the handles are rasped to provide comfort and efficiency to the user
- PC4.** bind the handle with traditional linen thread which is applied on a custom-made binding lather
- PC5.** operate foot treadle to bind the handle
- PC6.** apply appropriate glue to the binding and leave it for drying
- PC7.** polish the cricket bat once the bindings dry
- PC8.** use compression machine to ensure rubber, foam and plywood stuck together on a table tennis racquet

String and apply finishing touch to racquets

To be competent, the user/individual on the job must be able to:

- PC9.** use stringing machine to force the strings through the holes using threader
- PC10.** stress test the racquets to determine proper stiffness and hardness
- PC11.** inspect weight of racquets, usually before and after stringing, to make sure they meet specifications
- PC12.** check the balance of the racquets to ensure it is not too heavy at the head or at the handle
- PC13.** wound the grip smoothly ensuring no nicks or scratches
- PC14.** ensure the length-wise strings are threaded through forcefully first
- PC15.** use stringing machine to adjust the tension of the strings
- PC16.** ensure cross strings are threaded through afterward and the tension is adjusted
- PC17.** wrap handle in double sided tape and apply vinyl grip tap

Perform final quality and standards checks of bats and racquets

To be competent, the user/individual on the job must be able to:

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- PC18.** perform visual inspection to identify the faults, if any
- PC19.** transfer finished racquets to lab tests to ensure the rackets are within the given constraints
- PC20.** ensure the quality of semi-finished goods meet specifications
- PC21.** identify if the product conforms to shape and size requirements
- PC22.** ensure the shape, size and weight of the finished goods is as per the manufacturing standards
- PC23.** identify, mark and place rejects in the designated location
- PC24.** use tools and equipment correctly, efficiently and safely
- PC25.** complete and store accurate records and documentation
- PC26.** handover to alteration section to address minor defects
- PC27.** attach brand labels, barcodes/price tags on the article
- PC28.** clean the final article to remove all residue, dirt, adhesive or any other contamination by hand/ machine
- PC29.** wrap the article in paper and prepare for packing in boxes

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC30.** interact (verbal, non-verbal and written) in a gender, disability, and culturally sensitive manner
- PC31.** promote a safe and interactive environment
- PC32.** identify and report inappropriate behaviour (e.g. sexual harassment) to appropriate authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the process for finishing
- KU2.** types of tools and equipment should be used for finishing
- KU3.** types of raw material should be used require for finishing
- KU4.** International standards of equipment
- KU5.** the Standard applicable for manufacturing a Cricket Ball of International Label
- KU6.** follow and maintain the BIS and International standards during process of cricket ball manufacturing
- KU7.** explain different inspection/ testing procedures necessary to produces a Cricket Ball of International Label
- KU8.** the process for packaging
- KU9.** types of tools and equipment and other materials used for packaging
- KU10.** the process of packaging in standard manner to prepare to send National and International Market
- KU11.** potential faults in the various processes and methods to avoid them
- KU12.** process for obtaining replacements for worn, faulty or defective tools
- KU13.** appearance of the final product
- KU14.** common faults in tools and equipment and implications of working with faulty equipment

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- KU15.** the tools, equipment and settings are needed for the operation
- KU16.** manufacturers' instructions
- KU17.** polishing, painting, shining different materials
- KU18.** plan an appropriate response after assessing an emergency situation
- KU19.** emergency response skills appropriate to the location and personal health care
- KU20.** importance of gender and its related concepts such as gender roles, gender equality, gender power relations etc.
- KU21.** POSH (Prevention of Sexual Harassment) Act

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** resolve conflicts
- GS2.** report and record incidents and emergency situations
- GS3.** read and understand all usage and safety manuals
- GS4.** write in English/ local language as applicable
- GS5.** fill up activity logs/work sheet in the prescribed format of the company as applicable
- GS6.** listen actively
- GS7.** communicate effectively with operators, supervisors, managers, etc

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assemble different components of bats</i>	8	20	-	-
PC1. identify the points of inspection of the assembling goods	1	3	-	-
PC2. rasp the handles using appropriate tools	1	3	-	-
PC3. ensure the handles are rasped to provide comfort and efficiency to the user	1	3	-	-
PC4. bind the handle with traditional linen thread which is applied on a custom-made binding lather	1	3	-	-
PC5. operate foot treadle to bind the handle	1	2	-	-
PC6. apply appropriate glue to the binding and leave it for drying	1	2	-	-
PC7. polish the cricket bat once the bindings dry	1	2	-	-
PC8. use compression machine to ensure rubber, foam and plywood stuck together on a table tennis racquet	1	2	-	-
<i>String and apply finishing touch to racquets</i>	9	18	-	-
PC9. use stringing machine to force the strings through the holes using threader	1	2	-	-
PC10. stress test the racquets to determine proper stiffness and hardness	1	2	-	-
PC11. inspect weight of racquets, usually before and after stringing, to make sure they meet specifications	1	2	-	-
PC12. check the balance of the racquets to ensure it is not too heavy at the head or at the handle	1	2	-	-
PC13. wound the grip smoothly ensuring no nicks or scratches	1	2	-	-
PC14. ensure the length-wise strings are threaded through forcefully first	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. use stringing machine to adjust the tension of the strings	1	2	-	-
PC16. ensure cross strings are threaded through afterward and the tension is adjusted	1	2	-	-
PC17. wrap handle in double sided tape and apply vinyl grip tap	1	2	-	-
<i>Perform final quality and standards checks of bats and racquets</i>	12	24	-	-
PC18. perform visual inspection to identify the faults, if any	1	2	-	-
PC19. transfer finished racquets to lab tests to ensure the rackets are within the given constraints	1	2	-	-
PC20. ensure the quality of semi-finished goods meet specifications	1	2	-	-
PC21. identify if the product conforms to shape and size requirements	1	2	-	-
PC22. ensure the shape, size and weight of the finished goods is as per the manufacturing standards	1	2	-	-
PC23. identify, mark and place rejects in the designated location	1	2	-	-
PC24. use tools and equipment correctly, efficiently and safely	1	2	-	-
PC25. complete and store accurate records and documentation	1	2	-	-
PC26. handover to alteration section to address minor defects	1	2	-	-
PC27. attach brand labels, barcodes/price tags on the article	1	2	-	-
PC28. clean the final article to remove all residue, dirt, adhesive or any other contamination by hand/ machine	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. wrap the article in paper and prepare for packing in boxes	1	2	-	-
<i>Work effectively with others</i>	3	6	-	-
PC30. interact (verbal, non-verbal and written) in a gender, disability, and culturally sensitive manner	1	2	-	-
PC31. promote a safe and interactive environment	1	2	-	-
PC32. identify and report inappropriate behaviour (e.g. sexual harassment) to appropriate authority	1	2	-	-
NOS Total	32	68	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SPF/N8105
NOS Name	Carry out final steps of crafting the bats and racquets
Sector	Sports
Sub-Sector	Sports Manufacturing
Occupation	Sports Manufacturing
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	NA
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

SPF/N1169: Improve workplace resource usage

Description

This unit is about increasing efficiencies and maximizing the use of materials, particularly resources, energy, and trash, in workplace operations.

Scope

The scope covers the following :

- conservation procedures for materials
- power conservation methods
- waste management/recycling procedures

Elements and Performance Criteria

conservation procedures for materials

To be competent, the user/individual on the job must be able to:

- PC1.** determine ways to optimize usage of material including water in various tasks/activities/processes
- PC2.** examine various tasks/activities/processes for spills/leaks
- PC3.** eliminate spills/leaks and escalate to appropriate authorities if they cannot be corrected
- PC4.** conduct routine cleaning of tools, machines, and equipment

power conservation practices

To be competent, the user/individual on the job must be able to:

- PC5.** determine ways to optimize usage of electricity/energy in various tasks/activities/processes
- PC6.** ensure if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC7.** report equipment malfunctions (fumes/sparks/emission/vibration/noise) and lapses in maintenance
- PC8.** ensure that all electrical equipment and appliances are correctly connected and turned off when not in use

waste management/recycling procedures

To be competent, the user/individual on the job must be able to:

- PC9.** determine recyclable and non-recyclable, and hazardous waste generated
- PC10.** separate waste into different categories
- PC11.** discard non-recyclable waste appropriately
- PC12.** store recyclable and reusable materials in a designated spot

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** potential hazards, risks and threats based on the nature of work
- KU2.** layout of the workstation and electrical and thermal equipment used
- KU3.** organizations' procedures for minimizing waste
- KU4.** efficient and inefficient utilization of material and water
- KU5.** ways of efficiently managing material and water in the process
- KU6.** basics of electricity and prevalent energy efficient devices
- KU7.** ways to recognize common electrical problems
- KU8.** common practices of conserving electricity
- KU9.** usage of different colours of dustbins
- KU10.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU11.** waste management and methods of waste disposal
- KU12.** common sources of pollution and ways to minimize it

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** read Standard Operating Practices (SOP) documents
- GS4.** communicate with colleagues on the significance of greening of jobs
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** work with supervisors/team members to carry out work related tasks
- GS8.** identify cause and effect of greening of jobs

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>conservation procedures for materials</i>	4	12	-	-
PC1. determine ways to optimize usage of material including water in various tasks/activities/processes	1	3	-	-
PC2. examine various tasks/activities/processes for spills/leaks	1	3	-	-
PC3. eliminate spills/leaks and escalate to appropriate authorities if they cannot be corrected	1	3	-	-
PC4. conduct routine cleaning of tools, machines, and equipment	1	3	-	-
<i>power conservation practices</i>	8	10	-	-
PC5. determine ways to optimize usage of electricity/energy in various tasks/activities/processes	2	3	-	-
PC6. ensure if the equipment/machine is functioning normally before commencing work and rectify wherever required	2	3	-	-
PC7. report equipment malfunctions (fumes/sparks/emission/vibration/noise) and lapses in maintenance	2	2	-	-
PC8. ensure that all electrical equipment and appliances are correctly connected and turned off when not in use	2	2	-	-
<i>waste management/recycling procedures</i>	8	8	-	-
PC9. determine recyclable and non-recyclable, and hazardous waste generated	2	2	-	-
PC10. separate waste into different categories	2	2	-	-
PC11. discard non-recyclable waste appropriately	2	2	-	-
PC12. store recyclable and reusable materials in a designated spot	2	2	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SPF/N1169
NOS Name	Improve workplace resource usage
Sector	Sports
Sub-Sector	Sports
Occupation	Sports Coaching
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 50% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SPF/N8103.Prepare tools and materials to make bats and racquets	20	80	0	0	100	20
SPF/N8104.Craft bats and racquets	54	146	0	0	200	40
SPF/N8105.Carry out final steps of crafting the bats and racquets	32	68	0	0	100	20
SPF/N1169.Improve workplace resource usage	20	30	0	0	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	146	354	0	0	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.